



VLSI Testing

積體電路測試

Logic Built-In Self Test (BIST)

*Part 2: ORA, BIST Architecture**

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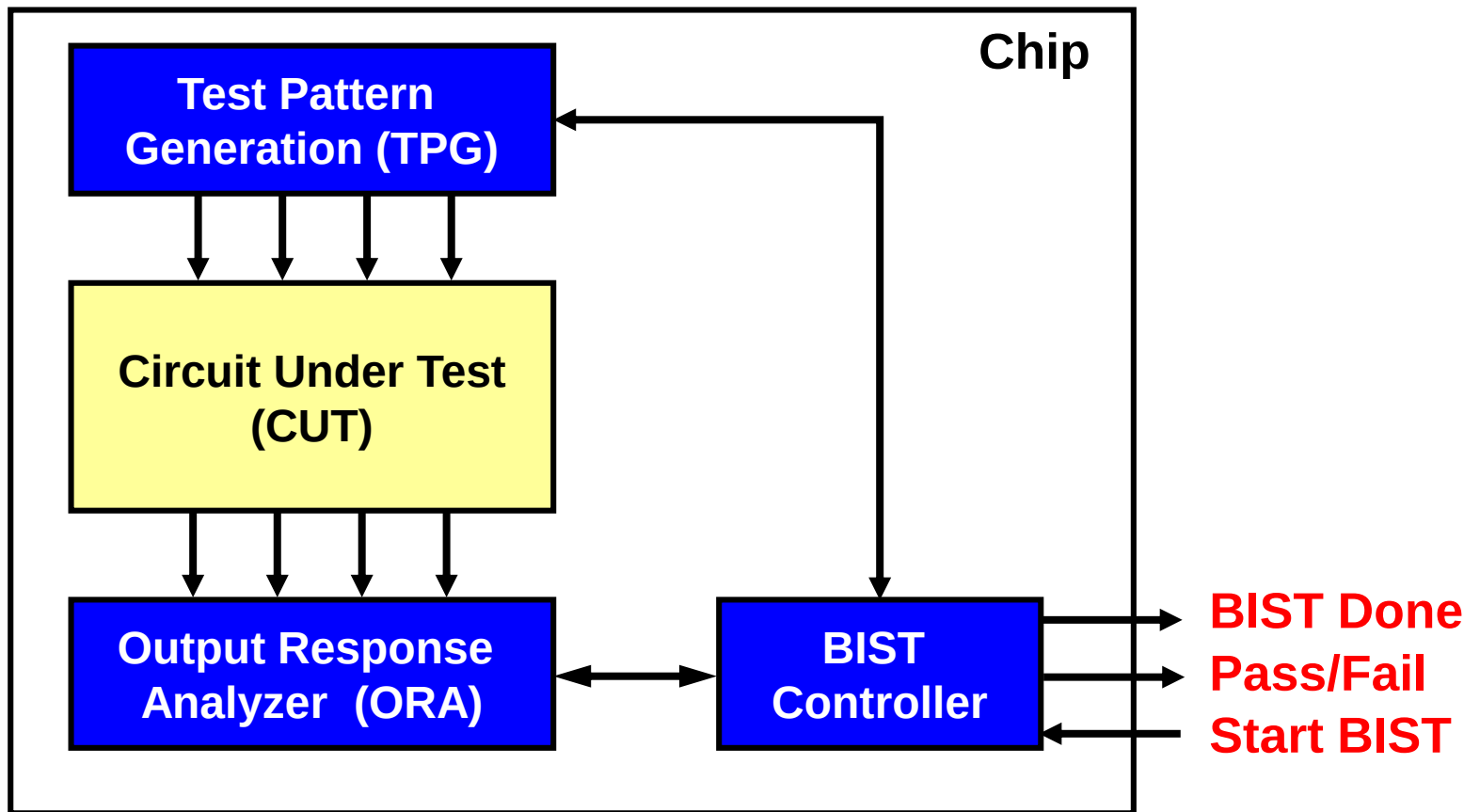
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* Some pictures are courtesy of Prof. Shi-yu Huang, NTHU

VLSI Test 14.1 © National Taiwan University

Architecture of BIST (Review)

- Three components: *BIST Controller, TPG, ORA*
- Three I/O Pins: *Start BIST, BIST Done, Pass/Fail*



Why Am I Learning This?

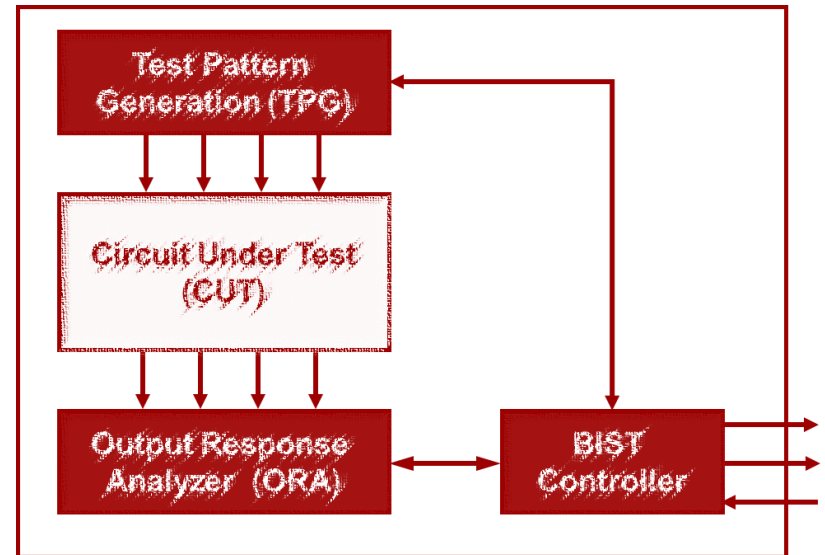
- **Output Response Analyzer**
 - ◆ Observe CUT output responses on chip
 - ◆ Very small area but correct Pass/Fail decision

*“What the superior man seeks is in himself;
what the small man seeks is in others.”*

君子求諸己，小人求諸人
(Confucius)

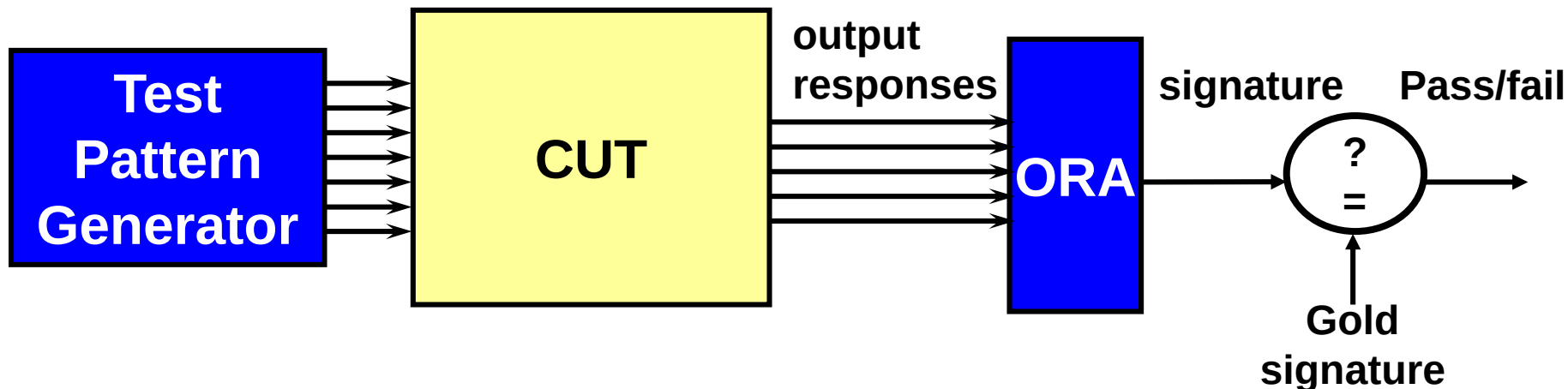
BIST Part 2

- Part 1
 - ◆ Introduction
 - ◆ Test Pattern Generation
- Part 2
 - ◆ Output Response Analysis
 - ◆ BIST Architecture
 - ◆ Issues with BIST
 - ◆ Conclusions



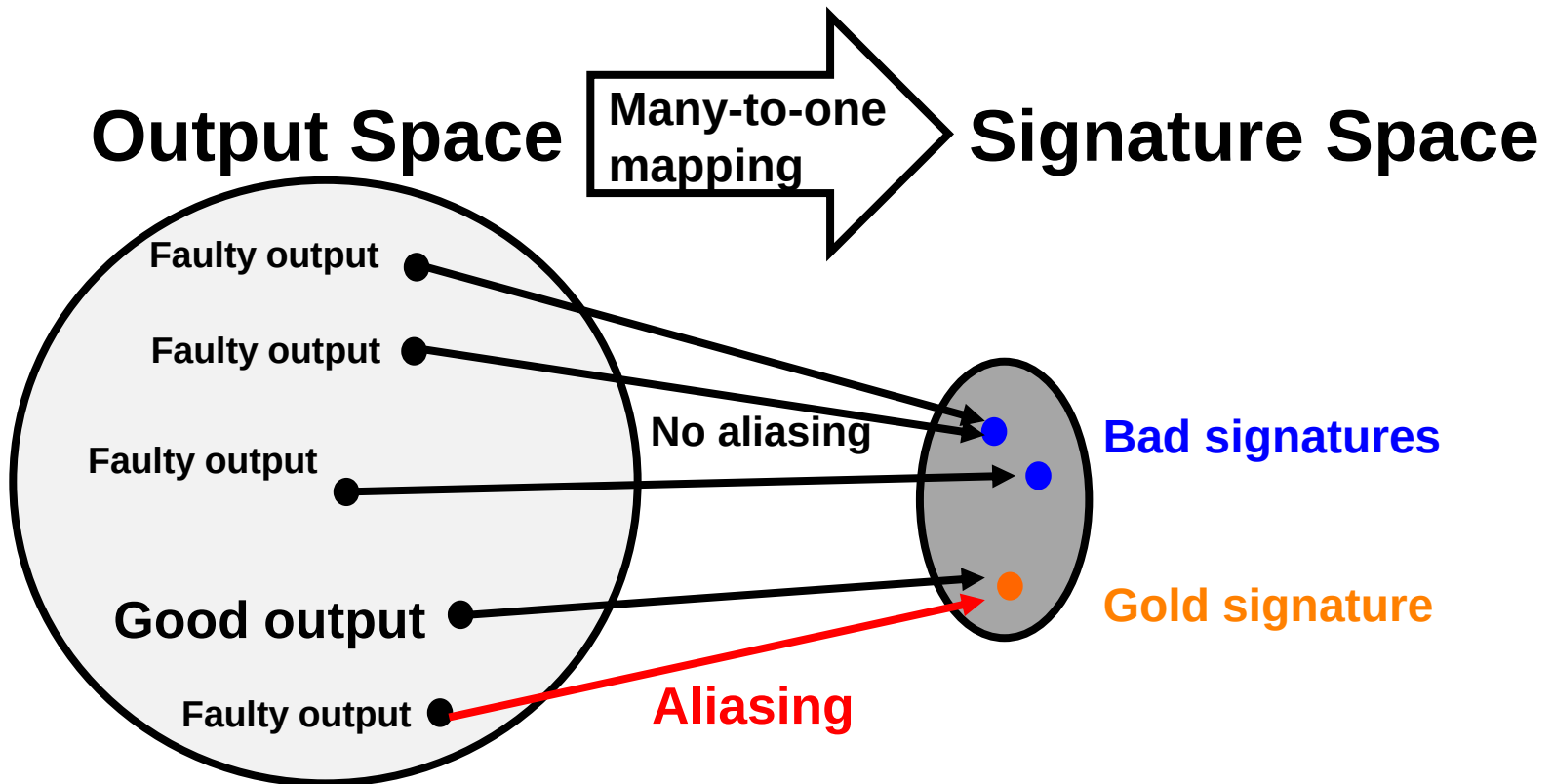
Output Response Analyzer

- What is ORA?
 - ♦ Compress CUT output responses into a small *signature*
 - ♦ Compare signature with *gold signature* to determine pass or fail
 - ♦ ORA also called *signature analyzer*
- What is good ORA? (very difficult to meet all requirements)
 - ♦ 1. Signature as *small* as possible
 - ♦ 2. *Correct Pass/Fail decision* (i.e. *low aliasing*, see next slide)
 - ♦ 3. *Small area*
 - ♦ 4. *Diagnosis support*



Aliasing

- **Aliasing** occurs when
 - ♦ $\text{signature}_{\text{faulty output}} = \text{signature}_{\text{good output}}$ (*gold signature*)
- Aliasing $\rightarrow$ Fault coverage loss $\rightarrow$ **Test escapes**
 - ♦ Defective circuits pass tests (**Very bad!**)



Probability of Aliasing

- Aliasing depends on many factors:
 - ♦ CUT, faults, test pattern, **ORA structure**
 - ♦ Exact analysis of aliasing difficult
 - ♦ Just **probability analysis** is good enough
- **Probability of Aliasing (PAL)**

$$PAL = \frac{\text{number of faulty outputs that generate gold signature}}{\text{total number of faulty outputs}}$$

- ♦ PAL between 0 and 1. Lower is better.

Good ORA Requires Low PAL

Quiz

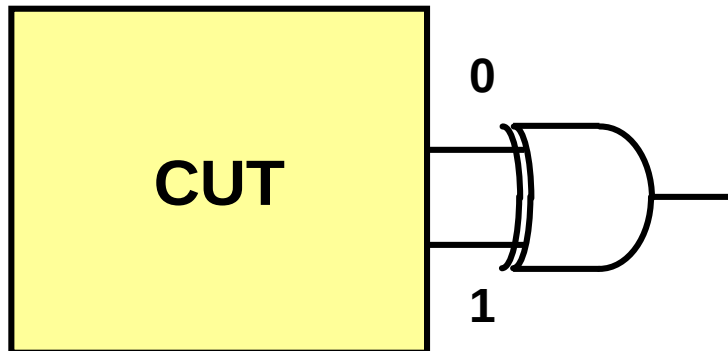
Q1: We use OR gate as ORA. Suppose good output is '01', what is gold signature?

ANS:

Q2: What is PAL of this ORA? Suppose '00, 01 10, 11' equally likely

$$PAL = \frac{\text{number of faulty outputs that generate gold signature}}{\text{total number of faulty outputs}}$$

ANS:



BIST Part 2

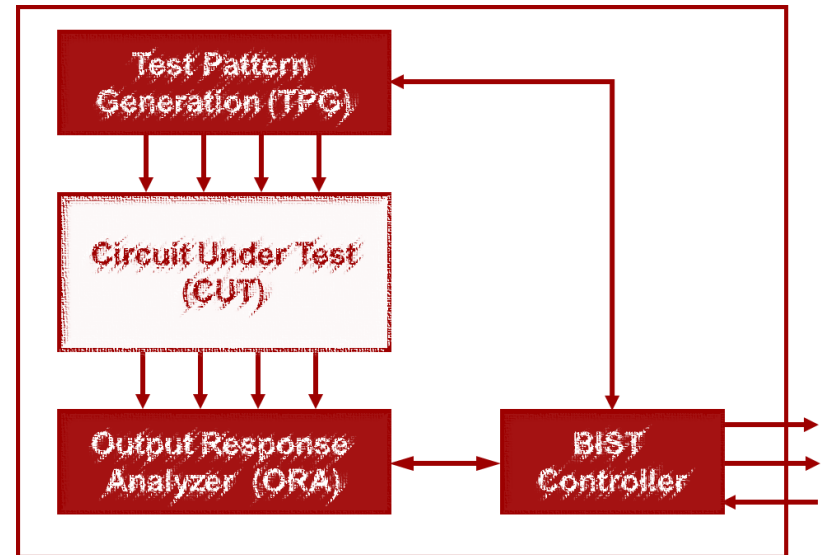
- Output Response Analysis

- ◆ Simple ORA

- * Duplication, Reverse operation
- * Checksum (aka. Parity Checker)
- * Ones counter
- * Transition counter

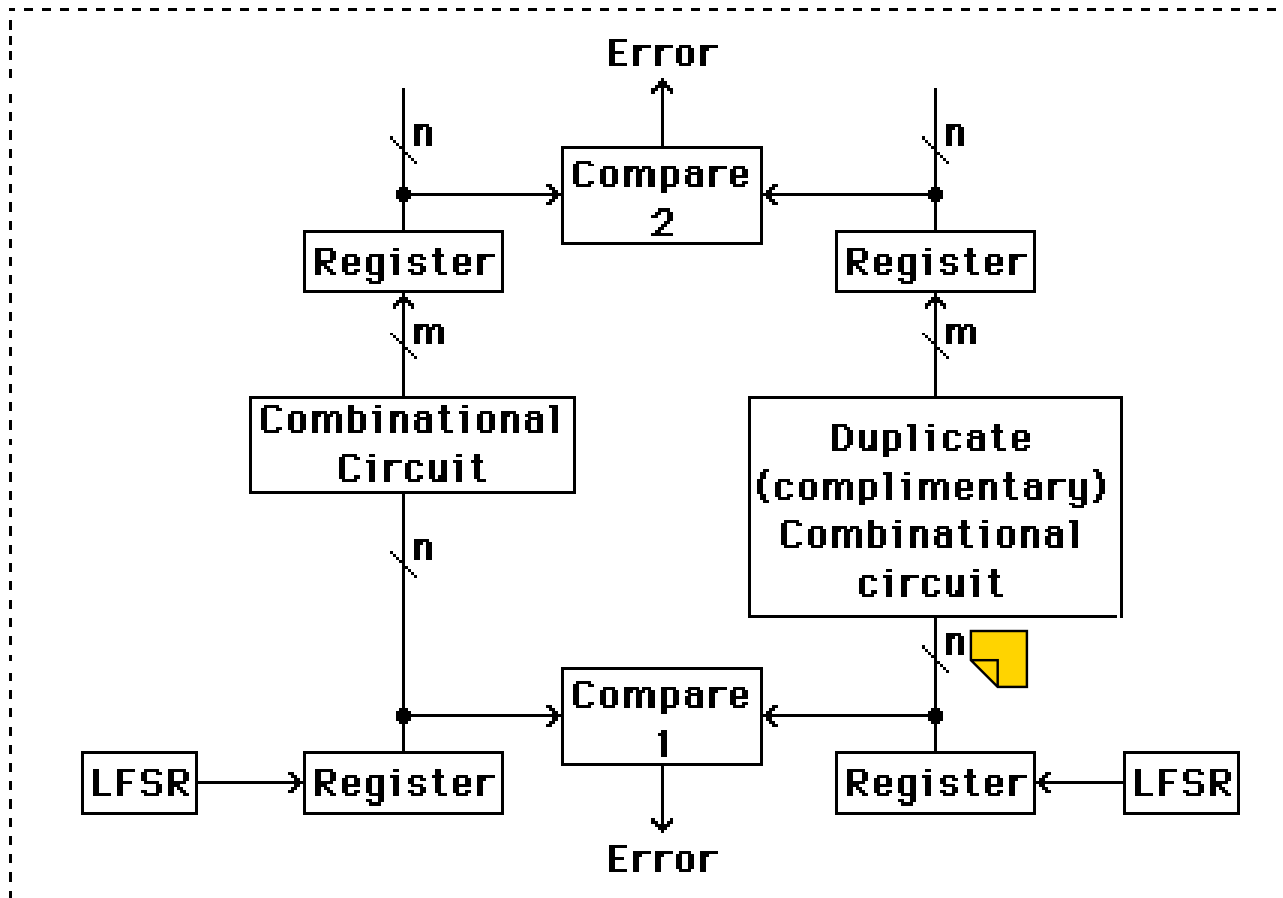
- ◆ LFSR-based

- BIST Architecture
- Issues with BIST
- Conclusions



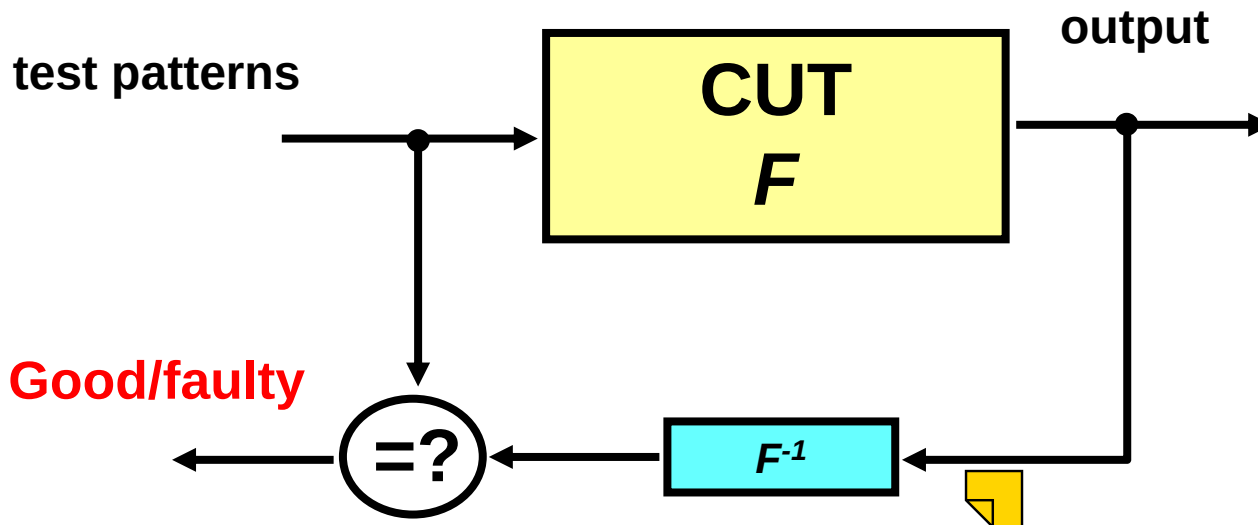
Duplication

- Advantage: low aliasing; good for on-line BIST
- Disadvantages: too big in area



Reverse Operation

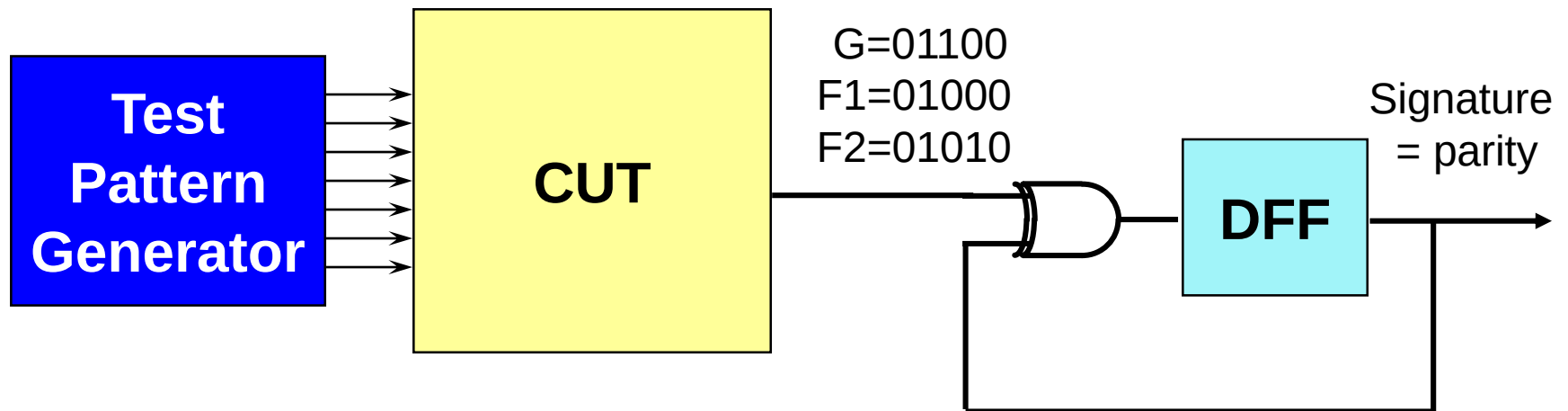
- Advantages
 - ♦ Applicable to both on-line and off-line BIST
 - ♦ Low aliasing
- Disadvantages
 - ♦ Not generally applicable to all CUT



F^{-1} is reverse operation of F

Parity Checker

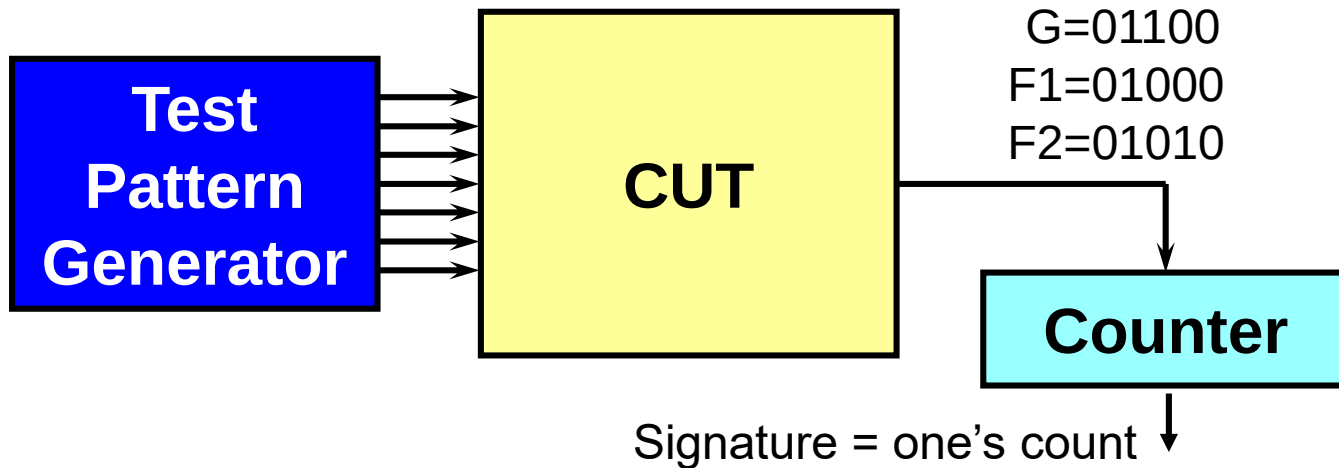
- Signature = Inversion parity of CUT output
 - ♦ Odd or even
- Example (initially DFF=0)
 - ♦ Good CUT output = 01100 → 0 gold signature
 - ♦ Faulty CUT1 output = 01000 → 1 detected, **no aliasing**
 - ♦ Faulty CUT2 output = 01010 → 0 not detected, **aliasing**



$$PAL = \frac{\text{number of faulty outputs that generate gold signature}}{\text{total number of faulty outputs}} = \frac{16-1}{2^5-1} = \frac{15}{31}$$

Ones Counter

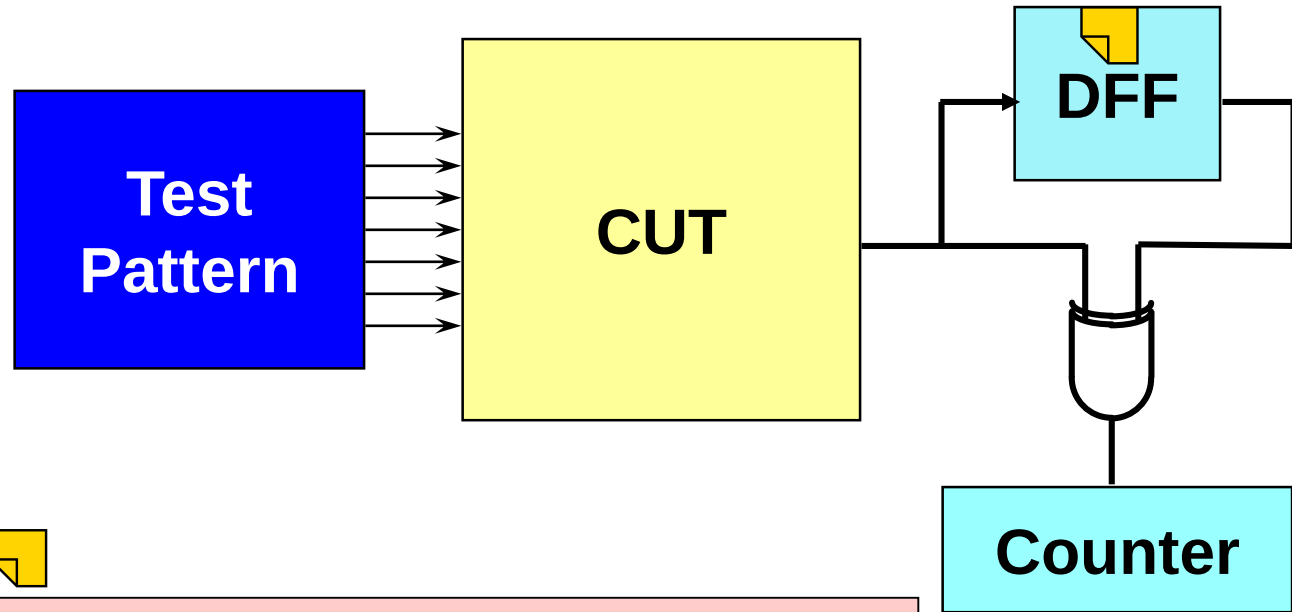
- Signature = number of ones
- Example: **sequence length=5=m; signature=2=n;**
 - ♦ Good CUT output = 01100 → 2 gold signature
 - ♦ Faulty CUT1 output = 01000 → 1 detected, **no aliasing**
 - ♦ Faulty CUT2 output = 01010 → 2 not detected, **aliasing**



$$PAL = \frac{C_n^m - 1}{2^m - 1} = \frac{C_2^5 - 1}{2^5 - 1} = \frac{10 - 1}{32 - 1} = \frac{9}{31}$$

Transition Counter

- Signature = number of **transitions** ($0 \rightarrow 1$, $1 \rightarrow 0$)
- Example: **sequence length**=5= m ; **signature**=2= n ;
 - ♦ Good CUT output = 01100 $\rightarrow$ 2
 - ♦ Faulty CUT1 output = 01000 $\rightarrow$ 2 not detected, **aliasing**
 - ♦ Faulty CUT2 output = 01010 $\rightarrow$ 4 detected, **no aliasing**



$$PAL = \frac{2C_n^{m-1} - 1}{2^m - 1} = \frac{2C_2^4 - 1}{2^5 - 1} = \frac{12 - 1}{32 - 1} = \frac{11}{31}$$

Summary

- Output Response Analyzer
 - ◆ Compress CUT output responses into **signatures**
 - ◆ Aliasing mean **faulty outputs compressed into gold signature**
 - * induces **test escape**
 - ◆ Probability of aliasing (**PAL**)

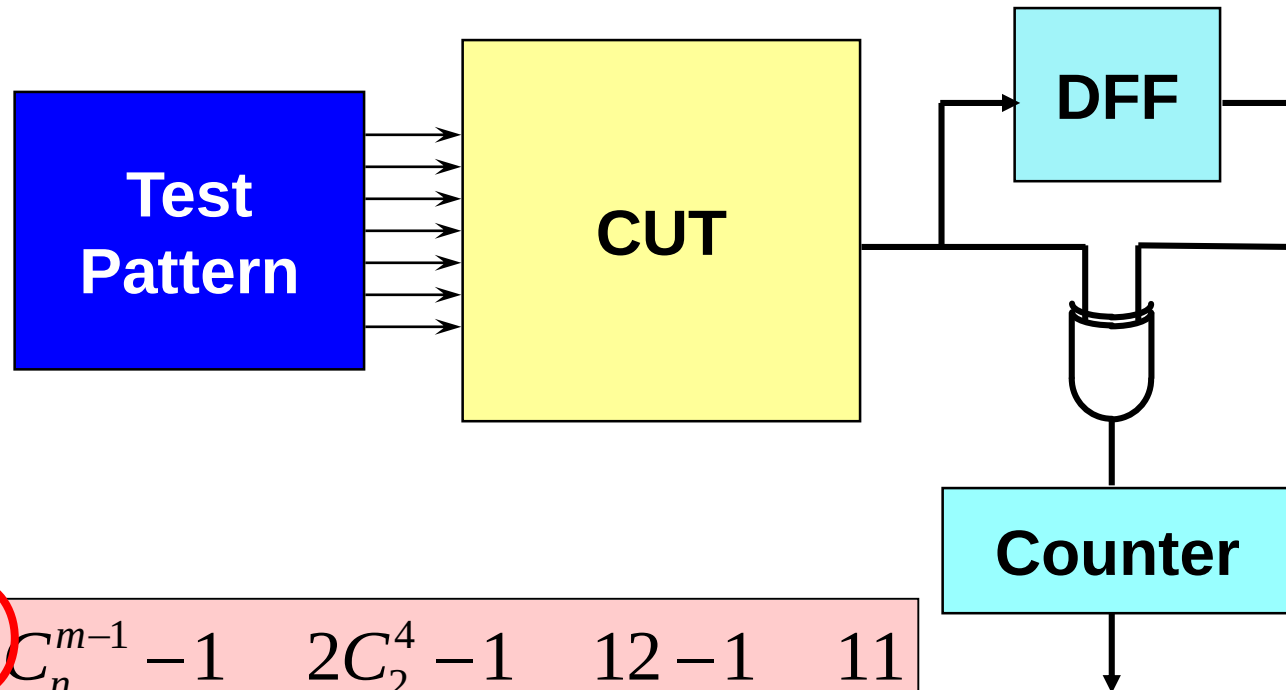
$$PAL = \frac{\text{number of faulty outputs that generate gold signature}}{\text{total number of faulty outputs}}$$

- ◆ Simple ORA
 - * Duplication
 - * Parity Checker
 - * Ones counter
 - * Transition counter

These ORA not Good Enough

FFT: Transition Counter

- Q: Why factor of 2 in PAL?



$$PAL = \frac{2C_n^{m-1} - 1}{2^m - 1} = \frac{2C_2^4 - 1}{2^5 - 1} = \frac{12 - 1}{32 - 1} = \frac{11}{31}$$